Dynamic analysis of beams, frames and grids

Special structural elements

ME473 Dynamic finite element analysis of structures

Stefano Burzio

2025



Where do we stand?

Week	Module	Lecture topic	Mini-projects
1	Linear elastodynamics	Strong and weak forms	
2		Galerkin and FE methods	
3		Systematisation of the procedure	Groups formation
4		3d elements, numerical integration	Project 1 statement
5	Special structural elements	Bars and trusses	
6		Beams, frames and grids	

Summary

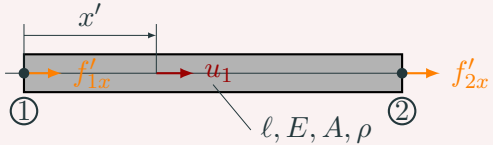
- Recap week 5
- Euler-Bernoulli planar beams
- Timoshenko planar beams
- Plane frames
- Plane grids
- Three-dimensional frames

Recommended readings

1. Logan, A first course in the finite element method, 6th ed. (chap. 4 and 5)
2. Paz and Leigh, Structural dynamics, 6th ed. (chap. 10, 11, 12 and 13)
3. Ferreira and Fantuzzi, MATLAB Codes for Finite Element Analysis, 2nd ed. (chap. 6, 7, 8, 9 and 10)

Recap week 5
Vibrations of bars and trusses

Non-oriented bar element



Differential equation governing the dynamics:

$$EA\partial^2_{x'x'}u_1(x',t) = \rho A\ddot{u}_1(x',t)$$

- Displacements approximation in local coordinate:

$$u_1^h(x',t) = \mathbf{H}(x')\mathbf{q}_{loc}(t) = \begin{bmatrix} h_1(x') & h_2(x') \end{bmatrix} \begin{bmatrix} q'_{1x}(t) \\ q'_{2x}(t) \end{bmatrix}$$

- Linear local shape functions:

$$h_1(x') = 1 - \frac{x'}{\ell} \qquad \text{and} \qquad h_2(x') = \frac{x'}{\ell}$$

- Semi-discrete weak form:

$$\boldsymbol{\delta}\mathbf{q}_{loc}^T\left[\mathbf{M}_{loc}\ddot{\mathbf{q}}_{loc}(t) + \mathbf{K}_{loc}\mathbf{q}_{loc}(t) - \mathbf{f}_{loc}(t)\right] = 0$$

Non-oriented bar element discretization

- Element stiffness matrix in local coordinates:

$$\mathbf{K}_{loc} = \int_0^\ell EA \, \frac{d\mathbf{H}}{dx'}^T \frac{d\mathbf{H}}{dx'} \, dx' = \int_0^\ell EA \begin{bmatrix} (h_1')^2 & h_1' h_2' \\ h_2' h_1' & (h_2')^2 \end{bmatrix} dx' = \frac{EA}{\ell} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

- Element consistent mass matrix in local coordinates:

$$\mathbf{M}_{loc} = \int_0^\ell \rho A \, \mathbf{H}^T \mathbf{H} \, dx' = \int_0^\ell \rho A \begin{bmatrix} (h_1)^2 & h_1 h_2 \\ h_2 h_1 & (h_2)^2 \end{bmatrix} dx' = \frac{\rho A \ell}{6} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

- Element applied loads vector in local coordinates:

$$\mathbf{f}_{loc}(t) = \begin{bmatrix} h_1(0) \\ h_2(0) \end{bmatrix} f_{1x}'(t) + \begin{bmatrix} h_1(\ell) \\ h_2(\ell) \end{bmatrix} f_{2x}'(t) = \begin{bmatrix} f_{1x}'(t) \\ f_{2x}'(t) \end{bmatrix}$$

Arbitrarly oriented bar element

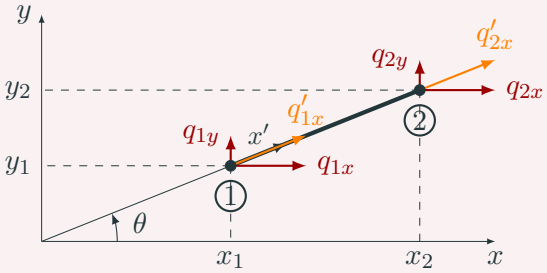
- Displacements in local coordinates:

$$\mathbf{q}_{loc} = [q'_{1x}, q'_{2x}]^T$$

- Displacements in global coordinates:

$$\mathbf{q} = [q_{1x}, q_{1y}, q_{2x}, q_{2y}]^T$$

- **Relation between local and global displacements:**



$$\underbrace{\begin{bmatrix} q'_{1x} \\ q'_{2x} \end{bmatrix}}_{\mathbf{q}_{loc}} = \underbrace{\begin{bmatrix} \cos(\theta) & \sin(\theta) & 0 & 0 \\ 0 & 0 & \cos(\theta) & \sin(\theta) \end{bmatrix}}_{\mathbf{T}} \underbrace{\begin{bmatrix} q_{1x} \\ q_{1y} \\ q_{2x} \\ q_{2y} \end{bmatrix}}_{\mathbf{q}}$$

Discretization of arbitrarly oriented bar

- Element stiffness matrix in global coordinates:

$$\mathbf{K} = \mathbf{T}^T \mathbf{K}_{loc} \mathbf{T} = \frac{EA}{\ell} \begin{bmatrix} \cos^2(\theta) & \sin(\theta)\cos(\theta) & -\cos^2(\theta) & -\sin(\theta)\cos(\theta) \\ & \sin^2(\theta) & -\sin(\theta)\cos(\theta) & -\sin^2(\theta) \\ & & \cos^2(\theta) & \sin(\theta)\cos(\theta) \\ \text{Symm.} & & & \sin^2(\theta) \end{bmatrix}$$

- Element consistent mass matrix in global coordinates:

$$\mathbf{M} = \mathbf{T}^T \mathbf{M}_{loc} \mathbf{T} = \frac{\rho A \ell}{6} \begin{bmatrix} 2\cos^2(\theta) & 2\sin(\theta)\cos(\theta) & \cos^2(\theta) & \sin(\theta)\cos(\theta) \\ & 2\sin^2(\theta) & \sin(\theta)\cos(\theta) & \sin^2(\theta) \\ & & 2\cos^2(\theta) & 2\sin(\theta)\cos(\theta) \\ \text{Symm.} & & & 2\sin^2(\theta) \end{bmatrix}$$

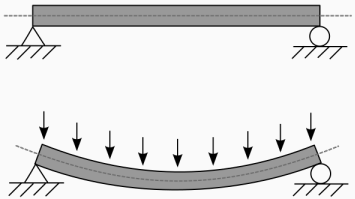
- Element applied loads vector in global coordinates:

$$\mathbf{f} = \mathbf{T}^T \mathbf{f}_{loc} = \begin{bmatrix} \cos(\theta) f_1 & \sin(\theta) f_1 & \cos(\theta) f_2 & \sin(\theta) f_2 \end{bmatrix}^T$$

Euler-Bernoulli planar beams

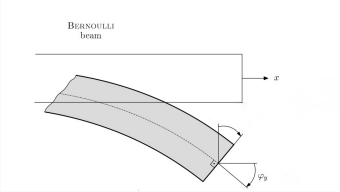
What is a beam?

- Considered to be a uniaxial (slender) element: the longitudinal direction is sufficiently larger than the other two.
- Cross-section does not change along the element's length.
- Subjected to **transversal loading** that produced significant bending effects.
- Very common type of structures used in steel buildings, bridges, towers, etc...



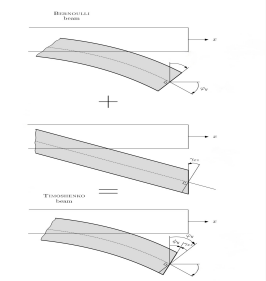
Euler-Bernoulli and Timoshenko beam assumptions

Thin beam theory



- Cross-sections remain **planar** and **perpendicular** to the longitudinal axis after bending.
- Shear deformation ε_{12} is neglected.
- The cross-section is infinitely rigid in its own plane.
- Suitable for thin beams made of isotropic materials.

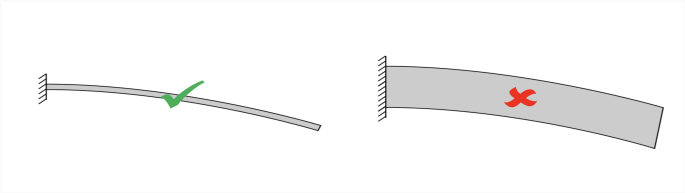
Thick beam theory



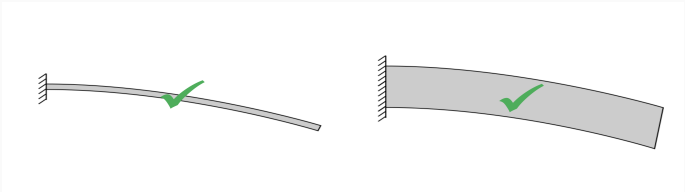
- Cross-sections remain **planar** but **not perpendicular** to the longitudinal axis.
- Shear deformation ε_{12} is considered.
- Suitable when shear effects matter, including in composite thick beams.

Validity range of beam theories

Euler–Bernoulli: valid for slender beams: $h/\ell < 1/100$



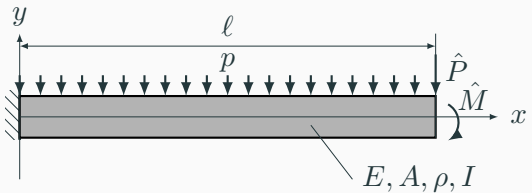
Timoshenko: valid for slender beams ($h/\ell < 1/100$) and thick beams ($h/\ell > 1/10$)



(Credit: Chatzi and Egger)

Kinematic assumptions for *planar* Euler-Bernoulli beam

1. The beam dynamics will be restricted to the $O(x, y)$ plane.
2. The beam cross-section remains planar after deformation.
3. Lines that are straight and perpendicular to the beam axis remain straight and perpendicular during deformation.



Model parameters:

- ℓ length
- A cross-sectional area
- E Young's modulus
- ρ material density
- I moment of inertia

Loads:

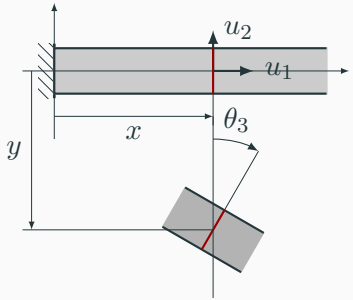
- $\hat{M}$ bending moment at free end
- $\hat{P}$ load at free end
- p distributed transversal load

Variables:

- $u_1(x, t)$ axial displacement
- $u_2(x, t)$ transversal displacement

Displacements field

Introduce an auxiliary variable $\theta_3(x,t)$ representing the total rotation of the section around the Oz axis. Rotations are positive in clockwise direction.



- The first Euler-Bernoulli assumption implies

$$u_3 = 0.$$

- The second Euler-Bernoulli assumption implies

$$u_1 = -y\theta_3 \qquad \textit{(rotation-axial displ.)}$$

- The third Euler-Bernoulli assumption implies

$$\theta_3 = \partial_x u_2 \qquad \textit{(rotation-transversal displ.)}$$

Only one unknown: transversal displacement $u_2(x,t)$.

Strain-displacement relationships

Substituting the displacement field $\mathbf{u}$ into $\boldsymbol{\varepsilon} = \boldsymbol{\nabla}\mathbf{u}$ yield to :

$$\varepsilon_{11} = \partial_x u_1 = -y\partial^2_{xx} u_2 \qquad \varepsilon_{22} = \partial_y u_2 = 0$$
$$\varepsilon_{12} = \partial_x u_2 + \partial_y u_1 = \partial_x u_2 - \theta_3 = 0 \qquad \varepsilon_{33} = \varepsilon_{23} = \varepsilon_{13} = 0$$

Generalized Hooke's law

Using the linear relationship for homogeneous and isotropic material, $\boldsymbol{\sigma} = \mathbf{C}\boldsymbol{\varepsilon}$, is possible to write the stress field in term of the strain field as :

$$\sigma_{11} = (\lambda + 2G)\varepsilon_{11} \qquad \sigma_{22} = \lambda\varepsilon_{11}$$
$$\sigma_{33} = \lambda\varepsilon_{11} \qquad \sigma_{12} = \sigma_{13} = \sigma_{23} = 0$$

where $\lambda = \frac{E}{(1+\nu)(1-2\nu)}$ and the shear modulus $G = \frac{E}{2(1+\nu)}$ are Lamé constants.

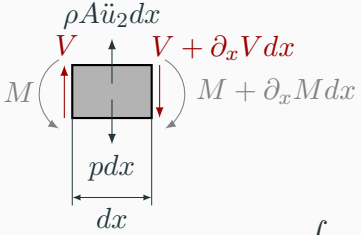
Dynamic equilibrium equations

The theoretical stresses do not agree well with the experimental measurements.

Additional assumptions regarding the stress field

$$\sigma_{11} = E\varepsilon_{11}, \qquad \text{and} \qquad \sigma_{22} = \sigma_{33} = \sigma_{12} = \sigma_{13} = \sigma_{23} = 0.$$

Justification : the dimension of the cross-section is much smaller that the beam length.



- Equilibrium of transverse forces:

$$\partial_x V + p = \rho A \ddot{u}_2$$

- Equilibrium of moments:

$$\partial_x M + V = 0$$

$$M = -\int_A y\sigma_{11}\, dA = -\int_A yE\varepsilon_{11}\, dA = EI\partial^2_{xx} u_2$$

Strong form for transversal vibrations for Euler-Bernoulli beam

The strong form consists of finding the transversal displacement $u_2 \in C^4([0,\ell]\times[0,T])$ such that the following equilibrium equation is satisfied:

$$\partial^2_{xx}\left(EI\partial^2_{xx}u_2\right) + \rho A\ddot{u}_2 = p.$$

Coupled with four boundary conditions:

$$u_2(0,t) = 0 \qquad \partial_x\left(-EI\partial^2_{xx}u_2\right)(\ell,t) = \hat{P}$$
$$\partial_x u_2(0,t) = 0 \qquad EI\partial^2_{xx}u_2(\ell,t) = \hat{M}$$

and two initial conditions:

$$u_2(x,0) = u_0(x) \qquad \dot{u}_2(x,0) = v_0(x)$$

Weak form for transversal vibrations for Euler-Bernoulli beam

Find the transversal displacement $u_2 \in \mathcal{U}$ such that the following equation is satisfied for every virtual transversal displacement $\delta u_2 \in \mathcal{V}$:

$$\int_0^\ell EI\, \partial_{xx}^2 u_2\, \partial_{xx}^2 \delta u_2\, dx + \int_0^\ell \rho A\, \ddot{u}_2\, \delta u_2\, dx = \int_0^\ell p\, \delta u_2\, dx + \hat{M}\, \delta u_2'(\ell) + \hat{P}\, \delta u_2(\ell)$$

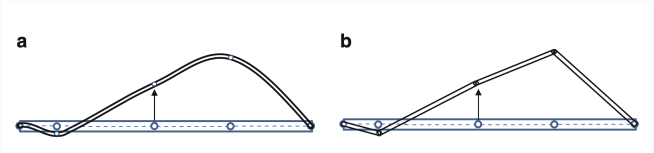
$$\mathcal{U} = \left\{ u_2(\cdot, t) \in H^2(]0, \ell[) \mid u_2(0,t) = \partial_x u_2(0,t) = 0 \ \forall t \in]0, T[\right\}$$
$$\mathcal{V} = \left\{ \delta u_2 \in H^2(]0, \ell[) \mid \delta u_2(0) = \delta u_2'(0) = 0 \right\}$$

The Sobolev space $H^2(]0, \ell[)$ is the defined as:

$$H^2(]0, \ell[) = \left\{ f \in L^2(]0, \ell[) \mid \int_0^\ell \left(\partial_x f(x)\right)^2 dx < \infty,\ \int_0^\ell \left(\partial_{xx}^2 f(x)\right)^2 dx < \infty \right\}.$$

Approximated transversal displacement

- Euler-Bernoulli theory states that both transversal displacement u_2 and rotation $\theta_3 = \partial_x u_2$ must be differentiable within finite elements and continuous between elements.
- Shape functions that meet this requirement are said to have **C^1 continuity**.

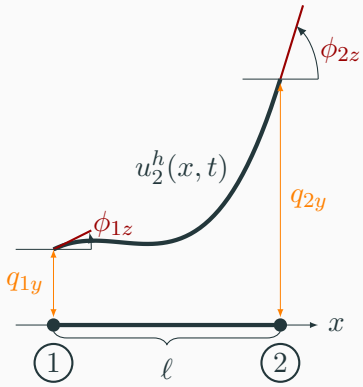


Cubic vs. linear elements. *Credit: [N]*

Assume the approximated transversal displacement to be

$$u_2^h(x,t) = a_3(t)x^3 + a_2(t)x^2 + a_1(t)x + a_0(t)$$

Nodal degrees of freedom within a beam element



Parameters shift:

$$a_0,\ a_1,\ a_2,\ a_3 \quad \Rightarrow \quad q_{1y},\ q_{2y},\ \phi_{1z},\ \phi_{2z}$$

A planar beam element has two DOFs per node:

- q_{1y} et q_{2y}: **nodal displacements** in the transverse direction.
- ϕ_{1z} et ϕ_{2z}: **nodal rotations** around the axis normal to the beam plane.

Hermite C^1 shape functions

- Expressing the approximated displacement u^h as a function of the nodal four DOFs yield to:

$$u_2^h(x,t) = \mathbf{H}(x)\mathbf{q}(t) = [h_1(x),\ h_2(x),\ h_3(x),\ h_4(x)] \begin{bmatrix} q_{1y}(t) \\ \phi_{1z}(t) \\ q_{2y}(t) \\ \phi_{2z}(t) \end{bmatrix}$$

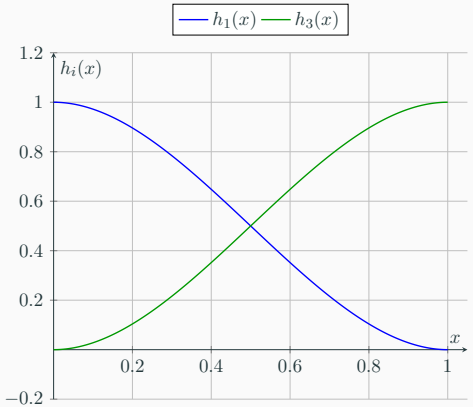
- **Cubic local shape functions:**

$$h_1(x) = 2(x/\ell)^3 - 3(x/\ell)^2 + 1 \qquad h_3(x) = 3(x/\ell)^2 - 2(x/\ell)^3$$
$$h_2(x) = x(1 - x/\ell)^2 \qquad h_4(x) = x(x/\ell)(x/\ell - 1)$$

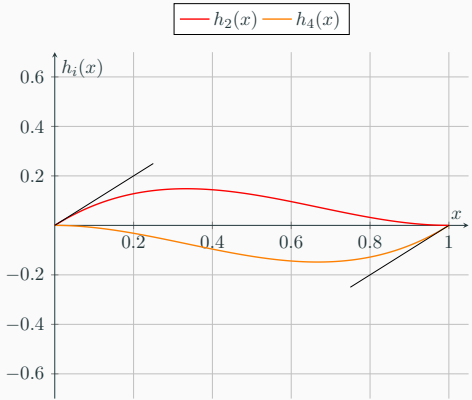
- Virtual displacement approximation follows the same logic:

$$\delta u^h(x) = \mathbf{H}(x)\boldsymbol{\delta}\mathbf{q} \qquad \text{where} \qquad \boldsymbol{\delta}\mathbf{q} = [\delta u_1,\ \delta\phi_{1z},\ \delta q_{2y},\ \delta\phi_{2z}]^T$$

Hermite C^1 shape functions visualizations



Translational shape functions



Rotational shape functions

Element stiffness matrix

$$\mathbf{K} = \int_0^\ell EI\ \mathbf{B}^T\mathbf{B}\ dx$$

$$= \int_0^\ell EI \begin{bmatrix} (h_1'')^2 & h_1''h_2'' & h_1''h_3'' & h_1''h_4'' \\ & (h_2'')^2 & h_2''h_3'' & h_2''h_4'' \\ & & (h_3'')^2 & h_3''h_4'' \\ Symm. & & & (h_4'')^2 \end{bmatrix} dx$$

$$= \frac{EI}{\ell^3} \begin{bmatrix} 12 & 6\ell & -12 & 6\ell \\ & 4\ell^2 & -6\ell & 2\ell^2 \\ & & 12 & -6\ell \\ Symm. & & & 4\ell^2 \end{bmatrix}$$

The deformation matrix is

$$\mathbf{B} = \frac{d^2\mathbf{H}}{dx^2} = \begin{bmatrix} h_1''(x) & h_2''(x) & h_3''(x) & h_4''(x) \end{bmatrix}$$

Element consistent mass matrix

$$\mathbf{M} = \int_0^\ell \rho A \, \mathbf{H}^T \mathbf{H} \, dx$$

$$= \int_0^\ell \rho A \begin{bmatrix} (h_1)^2 & h_1 h_2 & h_1 h_3 & h_1 h_4 \\ & (h_2)^2 & h_2 h_3 & h_2 h_4 \\ & & (h_3)^2 & h_3 h_4 \\ Symm. & & & (h_4)^2 \end{bmatrix} dx$$

$$= \frac{\rho A \ell}{420} \begin{bmatrix} 156 & 22\ell & 54 & -13\ell \\ & 4\ell^2 & 13\ell & -3\ell^2 \\ & & 156 & -22\ell \\ Symm. & & & 4\ell^2 \end{bmatrix}$$

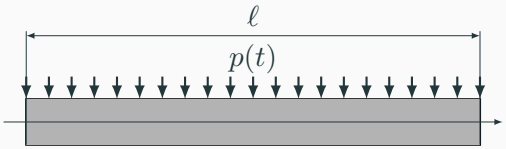
Element lumped mass matrix

- Assumption that the mass of the structure is lumped at the nodal coordinates where translational displacements are defined.
- Inertial effect associated with any rotational degree of freedom is usually assumed to be zero.
- Recall that $\rho\, A$ is the mass per unit length along the beam, then the lumped mass matrix is defined as

$$\mathbf{M} = \frac{\rho A \ell}{2} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Applied loads

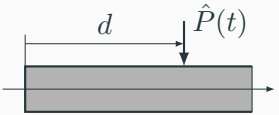
- **Uniformly distributed load:**



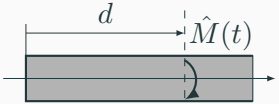
$$\mathbf{r} = \int_0^\ell p(t)\,\mathbf{H}^T\,dx = \frac{p(t)\ell}{2}\begin{bmatrix} 1 \\ \ell/6 \\ 1 \\ -\ell/6 \end{bmatrix}$$

An uniform transverse load can be replaced by two equivalent transverse nodes loads of value $(p\ell)/2$ and by two equivalent nodal moments of values $\pm p\ell^2/12$.

- **Concentrated loads:**



$$\mathbf{r}(t) = \hat{P}(t)\,\mathbf{H}^T(d)$$



$$\mathbf{r}(t) = \hat{M}(t)\,\frac{d\mathbf{H}}{dx}^T(d)$$

Post-processing: approximation of bending moment and shear force

- The approximated bending moment along the beam element is

$$M^h(x,t) = EI\partial^2_{xx}u^h_2(x,t) = EI\,\mathbf{B}(x)\mathbf{q}(t).$$

- The approximated shear force along the beam element is

$$V^h(x,t) = -\partial_x M^h(x,t) = -EI\,\frac{d\mathbf{B}}{dx}(x)\mathbf{q}(t).$$

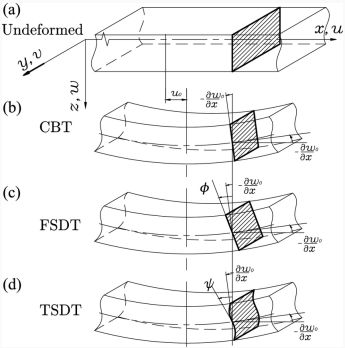
MATLAB examples

- A Cantilever beam subjected to a downward force
- A clamped-clamped beam (micro transducer) in free vibration.

Go to Matlab Drive

Timoshenko planar beam

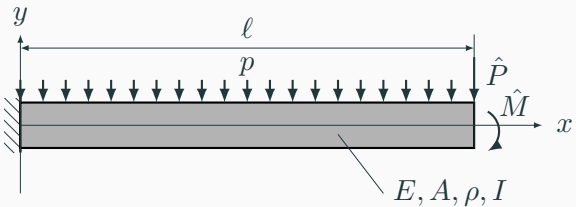
Higher-order deformation theories for beams



- **Classical beam theory** (Euler-Bernoulli) is a zero-order shear deformation theory.
- **Timoshenko theory** is a first-order shear deformation theory. Cross-sectional planes remain straight but not necessarily perpendicular to the mid-surface.
  - Displacements are linear functions through thickness.
  - In-plane strains are linear in z.
  - Shear strains are constant in z.
- **Third-order shear deformation theory**: normal cross-sectional planes to the mid-surface can rotate and deform.
  - Displacements are cubic functions through thickness.
  - In-plane strains are cubic in z.
  - Shear strains are quadratic in z.

Kinematic assumptions for planar Timoshenko beam

1. The beam dynamics will be restricted to the $O(x, y)$ plane.
2. The beam cross-section remains planar after deformation.
3. Lines that are straight and perpendicular to the geometrical beam axis remain straight but **not necessarily perpendicular** during deformation.



Model parameters:

- A cross-sectional area
- ℓ length
- E Young's modulus
- ρ material density
- I moment of inertia

Loads:

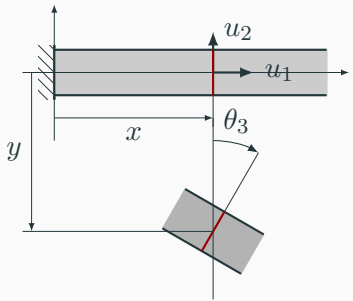
- $\hat{M}$ bending moment at free end
- $\hat{P}$ load at free end
- p distributed transversal load

Variables:

- $u_1(x, t)$ axial displacement
- $u_2(x, t)$ transversal displacement

Displacements field

Introduce an auxiliary variable $\theta_3(x,t)$ representing the total rotation of the section around the Oz axis. Rotations are positive in clockwise direction.



- The first Timoshenko assumption implies

$$u_3 = 0.$$

- The second Timoshenko assumption implies

$$u_1 = -y\theta_3 \qquad \textit{(rotation-axial displ.)}$$

Two unknowns: $u_2(x,t)$ transversal displacement and $\theta_3(x,t)$ rotation of the section around the Oz axis.

Strain-displacement relationships

Substituting the displacement field $\mathbf{u}$ into $\boldsymbol{\varepsilon} = \boldsymbol{\nabla}\mathbf{u}$ yield to :

$$\varepsilon_{11} = \partial_x u_1 = -y\partial^2_{xx} u_2 \qquad \varepsilon_{22} = \partial_y u_2$$
$$\varepsilon_{12} = \partial_x u_2 + \partial_y u_1 = \partial_x u_2 - \theta_3 \qquad \varepsilon_{33} = \varepsilon_{23} = \varepsilon_{13} = 0$$

Generalized Hooke's law

Using the linear relationship for homogeneous and isotropic material, $\boldsymbol{\sigma} = \mathbf{C}\boldsymbol{\varepsilon}$, is possible to write the stress field in term of the strain field as :

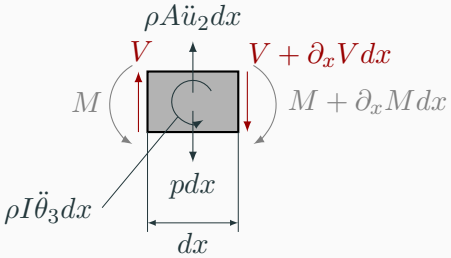
$$\sigma_{11} = (\lambda + 2G)\varepsilon_{11} \qquad \sigma_{12} = 2G\varepsilon_{12}$$
$$\sigma_{22} = \sigma_{33} = \lambda\varepsilon_{11} \qquad \sigma_{13} = \sigma_{23} = 0$$

where $\lambda = \frac{E}{(1+\nu)(1-2\nu)}$ and the shear modulus $G = \frac{E}{2(1+\nu)}$ are Lamé constants.

Dynamic equilibrium equations

Additional assumptions regarding the stress field

$$\sigma_{11} = E\varepsilon_{11}, \quad \sigma_{12} = 2G\varepsilon_{12} \qquad \text{and} \qquad \sigma_{22} = \sigma_{33} = \sigma_{13} = \sigma_{23} = 0.$$



- Equilibrium of transverse forces:

$$\partial_x V + p = \rho A\ddot{u}_2$$

- Equilibrium of moments:

$$\partial_x M + V = \rho I\ddot{\theta}_3$$

$$V = \int_A \sigma_{12}\, dA = kGA\varepsilon_{12} = kGA(\partial_x u_2 - \theta_3)$$

$$M = -\int_A y\sigma_{11}\, dA = -\int_A yE\varepsilon_{11}\, dA = EI\partial_x\theta_3$$

Strong form for transversal vibrations for Timoshenko beam

The weak formulation of the problem consists of finding the transversal displacement $u_2 \in C^2([0,\ell]\times[0,T])$ and the rotation $\theta_3 \in C^2([0,\ell]\times[0,T])$ such that the following equilibrium equations, boundary and initial conditions are satisfied.

$$\partial_x\big(kGA(\partial_x u_2 - \theta_3)\big) + p = \rho A \ddot{u}_2$$
$$\partial_x\big(EI\partial_x\theta_3\big) + kGA(\partial_x u_2 - \theta_3) = \rho I \ddot{\theta}_3$$

In matrix form: $\nabla_\sigma^T \mathbf{C} \nabla_u \mathbf{u} + \mathbf{f} = \mathbf{M}\ddot{\mathbf{u}}$, where

$$\underbrace{\begin{bmatrix}\partial_x & 0\\ 1 & \partial_x\end{bmatrix}}_{\nabla_\sigma^T}\underbrace{\begin{bmatrix}kGA & 0\\ 0 & EI\end{bmatrix}}_{\mathbf{C}}\underbrace{\begin{bmatrix}\partial_x & -1\\ 0 & \partial_x\end{bmatrix}}_{\nabla_u}\underbrace{\begin{bmatrix}u_2\\ \theta_3\end{bmatrix}}_{\mathbf{u}} + \underbrace{\begin{bmatrix}p\\ 0\end{bmatrix}}_{\mathbf{f}} = \underbrace{\begin{bmatrix}\rho A & 0\\ 0 & \rho I\end{bmatrix}}_{\mathbf{M}}\underbrace{\begin{bmatrix}\ddot{u}_2\\ \ddot{\theta}_3\end{bmatrix}}_{\ddot{\mathbf{u}}}$$

Boundary conditions

$$u_2(0,t) = 0 \qquad kGA\big(\partial_x u_2(\ell,t) - \theta_3(\ell,t)\big) = \hat{P}$$
$$\theta_3(0,t) = 0 \qquad EI\partial_x \theta_3(\ell,t) = \hat{M}$$

In matrix form: let $\hat{\mathbf{f}} = [\hat{P},\, \hat{M}]^T$ then

$$\mathbf{u}(0,t) = \mathbf{0}$$
$$\mathbf{C}\nabla_u \mathbf{u}(\ell,t) = \hat{\mathbf{f}}$$

Initial conditions

$$u_2(x,0) = u_0(x) \qquad \dot{u}_2(x,0) = v_0(x)$$
$$\theta_3(x,0) = \theta_0(x) \qquad \dot{\theta}_3(x,0) = \phi_0(x)$$

In matrix form: $\mathbf{u_0} = [u_0,\, \theta_0]^T$ and $\mathbf{v_0} = [v_0,\, \phi_0]^T$ then

$$\mathbf{u}(x,0) = \mathbf{u_0}$$
$$\dot{\mathbf{u}}(x,0) = \mathbf{v_0}$$

Weak form for transversal vibrations for Timoshenko beam

The weak formulation of the problem consists in finding the unknown function $\mathbf{u} \in \mathcal{U}$ which satisfies the equation $\forall \boldsymbol{\delta}\mathbf{u} \in \mathcal{V}$:

$$\int_0^\ell (\nabla_u \boldsymbol{\delta}\mathbf{u})^{\mathrm{T}}\, \mathbf{C}\, (\nabla_u \mathbf{u})\, \mathrm{d}x + \int_0^\ell \boldsymbol{\delta}\mathbf{u}^{\mathrm{T}} \mathbf{M}\ddot{\mathbf{u}}\, dx = \int_0^\ell \boldsymbol{\delta}\mathbf{u}^{\mathrm{T}} \mathbf{f} dx + \boldsymbol{\delta}\mathbf{u}^{\mathrm{T}}(\ell)\hat{\mathbf{f}}.$$

The function spaces $\mathcal{U}$ and $\mathcal{V}$ are defined as follows

$$\mathcal{U} = \left\{ \mathbf{u} = \{u_2, \theta_3\}^{\mathrm{T}} \mid u_2(\cdot, t) \in H^1(]0, \ell[); \theta_3(\cdot, t) \in H^1(]0, \ell[); u_2(0, t) = \theta_3(0, t) = 0 \right\},$$

$$\mathcal{V} = \left\{ \boldsymbol{\delta}\mathbf{u} = \{\delta u_2, \delta\theta_3\}^{\mathrm{T}} \mid \delta u_2 \in H^1(]0, \ell[); \delta\theta_3 \in H^1(]0, \ell[); \delta u_2(0) = \delta\theta_3(0) = 0 \right\}.$$

Displacement approximation

- We approximate the generalized displacement using the ansatz:

$$\mathbf{u}^h(\mathbf{x},t) = \mathbf{H}(\mathbf{x})\mathbf{q}(t) \qquad \text{and} \qquad \delta\mathbf{u}^h(\mathbf{x}) = \mathbf{H}(\mathbf{x})\delta\mathbf{q}$$

- **Cubic shape functions** are used to approximate the transversal displacement and **linear shape functions** for the rotation field:

$$\mathbf{H} = \begin{bmatrix} h_1^c & h_2^c & h_3^c & h_4^c \\ 0 & h_1^l & 0 & h_2^l \end{bmatrix}$$

$$h_1^c(x) = 2(x/\ell)^3 - 3(x/\ell)^2 + 1 \qquad h_3^c(x) = 3(x/\ell)^2 - 2(x/\ell)^3$$
$$h_2^c(x) = x(1 - x/\ell)^2 \qquad h_4^c(x) = x(x/\ell)(x/\ell - 1)$$
$$h_1^l(x) = 1 - x/\ell \qquad h_2^l(x) = x/\ell$$

Discretizations using linear shape functions can suffer from shear locking for thin beams !

- $\mathbf{q}(t)$ is a vector of (*unknown*) nodal displacements and $\boldsymbol{\delta}\mathbf{q}$ is a vector of constants:

$$\mathbf{q}(t) = \begin{bmatrix} q_{1y}(t) & \phi_{1z}(t) & q_{2y}(t) & \phi_{2z}(t) \end{bmatrix}^T \qquad \text{and} \qquad \boldsymbol{\delta}\mathbf{q} = \begin{bmatrix} \delta q_{1y} & \delta\phi_{1z} & \delta q_{2y} & \delta\phi_{2z} \end{bmatrix}^T$$

Deformation, stiffness, mass matrices and loads vector

- $\mathbf{B} = \nabla_u \mathbf{H}$ is the deformation matrix:

$$\mathbf{B} = \begin{bmatrix} \partial_x h_1^c & \partial_x h_2^c - h_1^l & \partial_x h_3^c & \partial_x h_4^c - h_2^l \\ 0 & \partial_x h_1^l & 0 & \partial_x h_2^l \end{bmatrix}$$

- The stiffness $\mathbf{K}$ and mass $\mathbf{Ma}$ matrices and load vector $\mathbf{r}$ are defined as

$$\mathbf{K} = \int_0^\ell \mathbf{B}^T \mathbf{C} \mathbf{B}\, dx,$$

$$\mathbf{Ma} = \int_0^\ell \mathbf{H}^T \mathbf{M} \mathbf{H}\, dx,$$

$$\mathbf{r}(t) = \int_0^\ell \mathbf{H}^T \mathbf{f}\, dx + \mathbf{H}^T(\ell)\hat{\mathbf{f}}.$$

- The semi-discrete equations of motion are given by

$$\mathbf{Ma}\,\ddot{\mathbf{q}}(t) + \mathbf{K}\mathbf{q}(t) = \mathbf{r}(t).$$

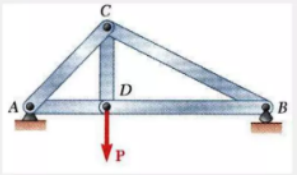
MATLAB example - modal analysis of a Timoshenko beam

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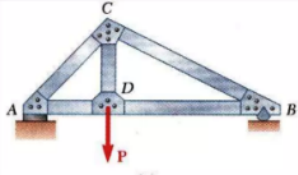
Planar frames

What is a plane frame?

- Structure composed of oriented thin beam elements, connected by welding, and carrying **transversal and axial forces** that all lies in a common plane.
- Both forces and moments can be transmitted between members.
- Loads are acting only in the common plane of the structure and they must be applied at the joints.



Plane truss

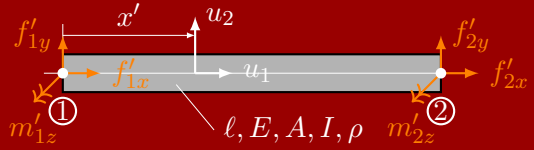


Plane frame

Truss + Thin beams = Frame

Axial loads in thin beam element

The inclusion of axial forces in a flexural beam element requires a *superposition* of bar and beam elements:



Differential equations governing the dynamics:

$$\partial^2_{x'x'}\left(EI\partial^2_{x'x'}u_2(x',t)\right) + \rho A\ddot{u}_2(x',t) = 0$$
$$EA\partial^2_{x'x'}u_1(x',t) = \rho A\ddot{u}_1(x',t)$$

Axial effects are subsequently incorporated into the beam element formulation, unless specified otherwise.

Axial displacement in thin beam element

A beam element now has three degrees of freedom per node: q'_{ix}, q'_{iy}, and ϕ'_{iz}.

- Displacements approximation:

$$u^h(x',t) = \mathbf{H}(x')\mathbf{q}_{loc}(t) = [h_1(x')\ h_2(x')\ h_3(x')\ h_4(x')\ h_5(x')\ h_6(x')] \begin{bmatrix} q'_{1x}(t) \\ q'_{1y}(t) \\ \phi'_{1z}(t) \\ q'_{2x}(t) \\ q'_{2y}(t) \\ \phi'_{2z}(t) \end{bmatrix}$$

- Local shape functions:

$$h_1(x') = 1 - x'/\ell \qquad h_4(x') = x'/\ell$$
$$h_2(x') = 2(x'/\ell)^3 - 3(x'/\ell)^2 + 1 \qquad h_5(x') = 3(x'/\ell)^2 - 2(x'/\ell)^3$$
$$h_3(x') = x'(1 - x'/\ell)^2 \qquad h_6(x') = x'(x'/\ell)(x'/\ell - 1)$$

Discretization of thin beam including axial effects

- Element stiffness matrix in local coordinates:

$$\mathbf{K}_{loc} = \frac{EI}{\ell^3} \begin{bmatrix} A\ell^2/I & 0 & 0 & -A\ell^2/I & 0 & 0 \\ & 12 & 6\ell & 0 & -12 & 6\ell \\ & & 4\ell^2 & 0 & -6\ell & 2\ell^2 \\ & & & A\ell^2/I & 0 & 0 \\ & & & & 12 & -6\ell \\ sym. & & & & & 4\ell^2 \end{bmatrix}$$

- Element consistent mass matrix in local coordinates:

$$\mathbf{M}_{loc} = \frac{\rho A\ell}{420} \begin{bmatrix} 140 & 0 & 0 & 70 & 0 & 0 \\ 0 & 156 & 22\ell & 0 & 54 & -13\ell \\ & & 4\ell^2 & 0 & 13\ell & -3\ell^2 \\ & & & 70 & 0 & 0 \\ & & & & 156 & -22\ell \\ sym. & & & & & 4\ell^2 \end{bmatrix}$$

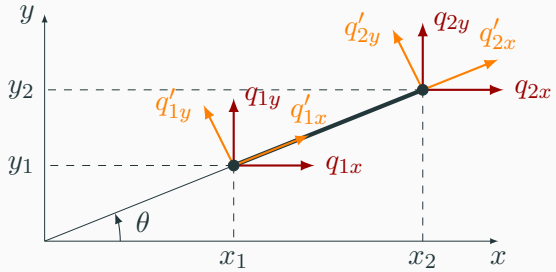
Arbitrarily oriented thin beam element including axial effects

- Displacements in local coordinates:

$$\mathbf{q}_{loc} = \begin{bmatrix} q'_{1x} & q'_{1y} & \phi'_{1z} & q'_{2x} & q'_{2y} & \phi'_{2z} \end{bmatrix}^T.$$

- Displacements in global coordinates:

$$\mathbf{q} = \begin{bmatrix} q_{1x} & q_{1y} & \phi_{1z} & q_{2x} & q_{2y} & \phi_{2z} \end{bmatrix}^T.$$



- **Relation between local and global displacements:**

$$\underbrace{\begin{bmatrix} q'_{1x} \\ q'_{1y} \\ \phi'_{1z} \\ q'_{2x} \\ q'_{2y} \\ \phi'_{2z} \end{bmatrix}}_{\mathbf{q}_{loc}} = \underbrace{\begin{bmatrix} \cos(\theta) & \sin(\theta) & 0 & 0 & 0 & 0 \\ -\sin(\theta) & \cos(\theta) & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \cos(\theta) & \sin(\theta) & 0 \\ 0 & 0 & 0 & -\sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}}_{\mathbf{T}} \underbrace{\begin{bmatrix} q_{1x} \\ q_{1y} \\ \phi_{1z} \\ q_{2x} \\ q_{2y} \\ \phi_{2z} \end{bmatrix}}_{\mathbf{q}}$$

Discretization of arbitrarly oriented beam including axial effects

- Element stiffness matrix in global coordinates:

$$\mathbf{K} = \mathbf{T}^T \mathbf{K}_{loc} \mathbf{T} = \frac{E}{\ell} \begin{bmatrix} \frac{A\ell^3C^2+12IS^2}{\ell^2} & \frac{(-12I+A\ell^2)CS}{\ell^2} & \frac{-6IS}{\ell} & -\frac{A\ell^3C^2+12IS^2}{\ell^2} & \frac{(12I-A\ell^2)CS}{\ell^2} & \frac{-6IS}{\ell} \\ & \frac{12IC^2+A\ell^2S^2}{\ell^2} & \frac{6IC}{\ell} & \frac{(12I-A\ell^2)CS}{\ell^2} & -\frac{12IC^2+A\ell^2S^2}{\ell^2} & \frac{6IC}{\ell} \\ & & 4I & \frac{6IS}{\ell} & \frac{-6IC}{\ell} & 2I \\ & & & \frac{A\ell^3C^2+12IS^2}{\ell^2} & \frac{(-12I+A\ell^2)CS}{\ell^2} & \frac{6IS}{\ell} \\ & & & & \frac{12IC^2+A\ell^2S^2}{\ell^2} & \frac{-6IC}{\ell} \\ & & & & & 4I \end{bmatrix}$$

where $C = \cos(\theta)$ and $S = \sin(\theta)$.

- Element consistent mass matrix in global coordinates:

$$\mathbf{M} = \mathbf{T}^T \mathbf{M}_{loc} \mathbf{T}$$

Applied loads

- Element applied loads vector in local coordinates:

$$\mathbf{f}_{loc}(t) = \begin{bmatrix} f'_{1x}(t) \\ f'_{1y}(t) \\ m'_{1z}(t) \\ f'_{2x}(t) \\ f'_{2y}(t) \\ m'_{2z}(t) \end{bmatrix}$$

- Element applied loads vector in global coordinates:

$$\mathbf{f} = \mathbf{T}^T \mathbf{f}_{loc} = \begin{bmatrix} \cos(\theta) f'_{1x} + \sin(\theta) f'_{1y} \\ -\sin(\theta) f'_{1x} + \cos(\theta) f'_{1y} \\ m'_{1z} \\ \cos(\theta) f'_{2x} + \sin(\theta) f'_{2y} \\ -\sin(\theta) f'_{2x} + \cos(\theta) f'_{2y} \\ m'_{2z} \end{bmatrix}$$

Assembly of stiffness and mass matrices and loads vector

Given a 2d frame structure made of m oriented beams, n nodes, and 3 DOFs per node:

1. **Element quantities:**
   - For each beam e, compute the element quantities global coordinates:

$$
\begin{aligned}
{}^{e}\mathbf{K} &= {}^{e}\mathbf{T}^{T}\,{}^{e}\mathbf{K}_{loc}\,{}^{e}\mathbf{T} \\
{}^{e}\mathbf{M} &= {}^{e}\mathbf{T}^{T}\,{}^{e}\mathbf{M}_{loc}\,{}^{e}\mathbf{T} \\
{}^{e}\mathbf{f} &= {}^{e}\mathbf{T}^{T}\,{}^{e}\mathbf{f}_{loc}
\end{aligned}
$$

2. **Global assembly:**
   - Initialize global stiffness matrix $\mathbf{K}$ and global mass matrix $\mathbf{M}$ of size $3n \times 3n$,
   - Initialize global loads vector $\mathbf{f}$ of size $3n \times 1$,
   - Assemble each ${}^{e}\mathbf{K}$, ${}^{e}\mathbf{M}$ and ${}^{e}\mathbf{f}$ for $e = 1, \ldots, m$, into $\mathbf{K}$, $\mathbf{M}$ and $\mathbf{f}$ respectively using element connectivity.

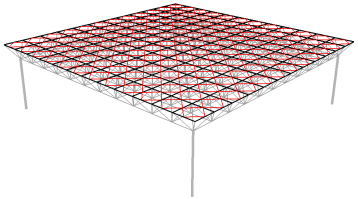
MATLAB example - frame 2d in free vibration

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Plane grids

What is a grid?

- A grid is a structure composed of oriented planar beams subjected to **perpendicular loading**, producing significant bending effects.
- Beams are connected by welding, so forces and moments can be transmitted between members.
- The axial effect of displacement is ignored for now.
- Grids commonly model floors, roofs, bridge decks, and similar structures.



Shaft + thin beams = Grid



Degrees of freedom identification

- **Planar frame under action of loads in the plane of the structure:**

  Components required to describe the displacements of joint i are:

  - q_{ix} translation in the x direction,
  - q_{iy} translation in the y direction,
  - ϕ_{iz} bending, rotation about the z axis.

- **Planar grids loaded perpendicularly to the plane of the structure:**

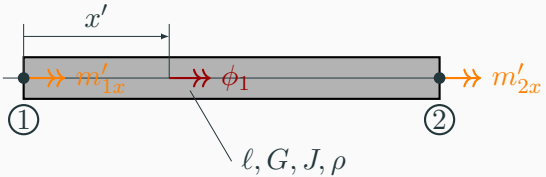
  Components required to describe the displacements of joint i are:

  - q_{iz} translation in the z direction,
  - ϕ_{ix} torsion, rotation about the x axis,
  - ϕ_{iy} bending, rotation about the z axis.

Shaft element

The development of a shaft finite element is very similar to the development of a bar finite element, where

- the axial displacement u_1 is replaced by the angular rotation ϕ_1,
- the axial nodal forces f'_{ix} are replaced by nodal torque m'_{ix},
- the element tensile stiffness AE/ℓ is replaced by the torsional stiffness GJ/ℓ.



- G shear modulus of the material
- J polar moment of inertia of the cross-section.
- ρ material density
- ℓ length
- ϕ_1 angular rotation
- x' (local) axial coordinate

Equation of motion for non-oriented shaft element

Differential equation governing the dynamics:

$$GJ\partial^2_{x'x'}\phi_1(x',t) = \rho J\ddot{\phi}_1(x',t)$$

- Angular rotation (twisting) approximation:

$$\phi_1^h(x',t) = \mathbf{H}(x')\mathbf{q}_{loc}(t) = \begin{bmatrix} h_1(x') & h_2(x') \end{bmatrix} \begin{bmatrix} \phi'_{1x}(t) \\ \phi'_{2x}(t) \end{bmatrix}$$

- Linear local shape functions:

$$h_1(x') = 1 - \frac{x'}{\ell} \qquad \text{and} \qquad h_2(x') = \frac{x'}{\ell}$$

- Semi-discrete weak form:

$$\boldsymbol{\delta}\mathbf{q}_{loc}^T\left[\mathbf{M}_{loc}\ddot{\mathbf{q}}_{loc}(t) + \mathbf{K}_{loc}\mathbf{q}_{loc}(t) - \mathbf{f}_{loc}(t)\right] = 0$$

Discretization of shaft

- Element stiffness matrix in local coordinates:

$$\mathbf{K}_{loc} = \int_0^\ell GJ \, \frac{d\mathbf{H}}{dx'}^T \frac{d\mathbf{H}}{dx'} \, dx' = \int_0^\ell GJ \begin{bmatrix} (h_1')^2 & h_1' h_2' \\ h_2' h_1' & (h_2')^2 \end{bmatrix} dx' = \frac{GJ}{\ell} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

- Element consistent mass matrix in local coordinates:

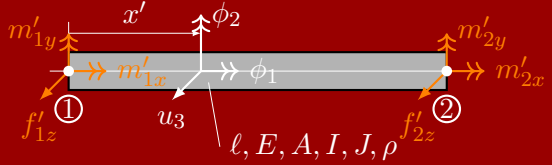
$$\mathbf{M}_{loc} = \int_0^\ell \rho J \, \mathbf{H}^T \mathbf{H} \, dx' = \int_0^\ell \rho J \begin{bmatrix} (h_1)^2 & h_1 h_2 \\ h_2 h_1 & (h_2)^2 \end{bmatrix} dx' = \frac{\rho J \ell}{6} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

- Element applied loads vector in local coordinates:

$$\mathbf{f}_{loc}(t) = \begin{bmatrix} h_1(0) \\ h_2(0) \end{bmatrix} m_{1x}'(t) + \begin{bmatrix} h_1(\ell) \\ h_2(\ell) \end{bmatrix} m_{2x}'(t) = \begin{bmatrix} m_{1x}'(t) \\ m_{2x}'(t) \end{bmatrix}$$

Torsional effects in thin beam element

The inclusion of torsional stiffness in a thin beam element, to model a typical element in a planar grid frame, requires a *superposition* of shaft and beam elements:



Differential equations governing the dynamics:

$$\partial^2_{x'x'}\left(EI\partial^2_{x'x'}u_3(x',t)\right) + \rho A\ddot{u}_3(x',t) = 0$$
$$GJ\partial^2_{x'x'}\phi_1(x',t) = \rho J\ddot{\phi}_1(x',t)$$

Torsional effects in thin beam element

Each grid element now has three degrees of freedom per node: ϕ'_{ix}, ϕ'_{iy} and q'_{iz}.

- Displacements approximation:

$$u^h(x',t) = \mathbf{H}(x')\mathbf{q}_{loc}(t) = [h_1(x')\ h_2(x')\ h_3(x')\ h_4(x')\ h_5(x')\ h_6(x')] \begin{bmatrix} \phi'_{1x}(t) \\ \phi'_{1y}(t) \\ q'_{1z}(t) \\ \phi'_{2x}(t) \\ \phi'_{2y}(t) \\ q'_{2z}(t) \end{bmatrix}$$

- Local shape functions:

$$h_1(x') = 1 - x'/\ell \qquad h_4(x') = x'/\ell$$
$$h_2(x') = x'(1 - x'/\ell)^2 \qquad h_5(x') = x'(x'/\ell)(x'/\ell - 1)$$
$$h_3(x') = 2(x'/\ell)^3 - 3(x'/\ell)^2 + 1 \qquad h_6(x') = 3(x'/\ell)^2 - 2(x'/\ell)^3$$

Discretization of thin beam with torsional effects

- Element stiffness matrix in local coordinates:

$$\mathbf{K}_{loc} = \frac{EI}{\ell^3}\begin{bmatrix} GJ\ell^2/EI & 0 & 0 & -GJ\ell^2/EI & 0 & 0 \\ & 4\ell^2 & -6\ell & 0 & 2\ell^2 & 6\ell \\ & & 12 & 0 & -6\ell & -12 \\ & & & GJ\ell^2/EI & 0 & 0 \\ & & & & 4\ell^2 & 6\ell \\ sym. & & & & & 12 \end{bmatrix}$$

- Element consistent mass matrix in local coordinates:

$$\mathbf{M}_{loc} = \frac{\rho A\ell}{420}\begin{bmatrix} 140J/A & 0 & 0 & 70J/A & 0 & 0 \\ 0 & 4\ell^2 & 22\ell & 0 & -3\ell^2 & 13\ell \\ & & 156 & 0 & -13\ell & 54 \\ & & & 140J/A & 0 & 0 \\ & & & & 4\ell^2 & -22\ell \\ sym. & & & & & 156 \end{bmatrix}$$

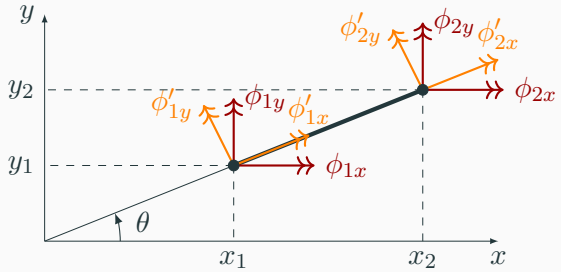
Arbitrarly oriented thin beam element with torsional effects

- Displacements in local coordinates:

$$\mathbf{q}_{loc} = \begin{bmatrix} \phi'_{1x} & \phi'_{1y} & q'_{1z} & \phi'_{2x} & \phi'_{2y} & q'_{2z} \end{bmatrix}^T .$$

- Displacements in global coordinates:

$$\mathbf{q} = \begin{bmatrix} \phi_{1x} & \phi_{1y} & q_{1z} & \phi_{2x} & \phi_{2y} & q_{2z} \end{bmatrix}^T .$$



- **Relation between local and global displacements:**

$$\underbrace{\begin{bmatrix} \phi'_{1x} \\ \phi'_{1y} \\ q'_{1z} \\ \phi'_{2x} \\ \phi'_{2y} \\ q'_{2z} \end{bmatrix}}_{\mathbf{q}_{loc}} = \underbrace{\begin{bmatrix} \cos(\theta) & \sin(\theta) & 0 & 0 & 0 & 0 \\ -\sin(\theta) & \cos(\theta) & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \cos(\theta) & \sin(\theta) & 0 \\ 0 & 0 & 0 & -\sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}}_{\mathbf{T}} \underbrace{\begin{bmatrix} \phi_{1x} \\ \phi_{1y} \\ q_{1z} \\ \phi_{2x} \\ \phi_{2y} \\ q_{2z} \end{bmatrix}}_{\mathbf{q}}$$

Discretization of arbitrarly oriented thin beam

- Element stiffness matrix in global coordinates:

$$\mathbf{K} = \mathbf{T}^T \mathbf{K}_{loc} \mathbf{T}$$

  where $C = \cos(\theta)$ and $S = \sin(\theta)$.
- Element consistent mass matrix in global coordinates:

$$\mathbf{M} = \mathbf{T}^T \mathbf{M}_{loc} \mathbf{T}$$

- Element applied loads vector in global coordinates:

$$\mathbf{f} = \mathbf{T}^T \mathbf{f}_{loc}$$

MATLAB example - dynamic response of a 2d grid

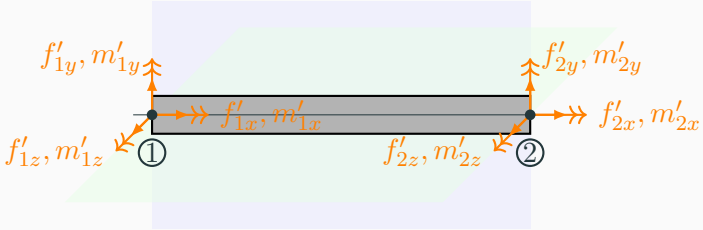
▸ Go to Matlab Drive

Three-dimensional frames

Example of a three-dimensional structure composed of thin beams



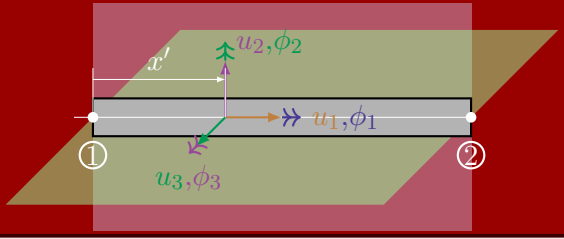
Credit: [N]



Three-dimensional thin beams are uniaxial (slender) element that can support:

- axial loads f'_{ix},
- torsional loads m'_{ix},
- bending in the $x' - y'$ plane: f'_{iy} and m'_{iz},
- bending in the $x' - z'$ plane: f'_{iz} and m'_{iy}.

Differential equations governing the dynamics of thin 3d beams



$$EA\partial^2_{x'x'}u_1(x',t) = \rho A\ddot{u}_1(x',t) \qquad \partial^2_{x'x'}\left(EI_z\partial^2_{x'x'}u_2(x',t)\right) + \rho A\ddot{u}_2(x',t) = 0$$

$$GJ\partial^2_{x'x'}\phi_1(x',t) = \rho J\ddot{\phi}_1(x',t) \qquad \partial^2_{x'x'}\left(EI_y\partial^2_{x'x'}u_3(x',t)\right) + \rho A\ddot{u}_3(x',t) = 0$$

- E Young's modulus of the material.
- A cross-sectional area.
- ρ material density.
- I_y and I_z are the second moments of inertia with respect to Oy and Oz.
- J polar moment of inertia of the cross-section.

Displacements discretization

Total of **six nodal generalized displacements at each unconstrained joint**:

- three translation components q'_{ix}, q'_{iy} and q'_{iz} along the x, y and z axes,
- three rotational components ϕ'_{ix}, ϕ'_{iy} and ϕ'_{iz} about the x, y and z axes.

$$u^h(x',t) = \mathbf{H}(x')\mathbf{q}_{loc}(t)$$

$$\mathbf{q}_{loc}(t) = \begin{bmatrix} q'_{1x}(t) \\ q'_{1y}(t) \\ q'_{1z}(t) \\ \phi'_{1x}(t) \\ \phi'_{1y}(t) \\ \phi'_{1z}(t) \\ q'_{2x}(t) \\ q'_{2y}(t) \\ q'_{2z}(t) \\ \phi'_{2x}(t) \\ \phi'_{2y}(t) \\ \phi'_{2z}(t) \end{bmatrix}$$

$$h_1(x') = 1 - x'/\ell$$
$$h_2(x') = 2(x'/\ell)^3 - 3(x'/\ell)^2 + 1$$
$$h_3(x') = 2(x'/\ell)^3 - 3(x'/\ell)^2 + 1$$
$$h_4(x') = 1 - x'/\ell$$
$$h_5(x') = x'(1 - x'/\ell)^2$$
$$h_6(x') = x'(1 - x'/\ell)^2$$
$$h_7(x') = x'/\ell$$
$$h_8(x') = 3(x'/\ell)^2 - 2(x'/\ell)^3$$
$$h_9(x') = 3(x'/\ell)^2 - 2(x'/\ell)^3$$
$$h_{10}(x') = x'/\ell$$
$$h_{11}(x') = x'(x'/\ell)(x'/\ell - 1)$$
$$h_{12}(x') = x'(x'/\ell)(x'/\ell - 1)$$

Element stiffness matrix in local coordinates

The stiffness matrix for a three-dimensional uniform beam element is written as the superposition of the **axial stiffness** matrix, the **torsional stiffness** matrix and the **flexural stiffness** matrix:

$$\mathbf{K}_{loc} = \begin{bmatrix} \frac{EA}{\ell} & 0 & 0 & 0 & 0 & 0 & -\frac{EA}{\ell} & 0 & 0 & 0 & 0 & 0 \\ & \frac{12EI_z}{\ell^3} & 0 & 0 & 0 & \frac{6EI_z}{\ell^2} & 0 & -\frac{12EI_z}{\ell^3} & 0 & 0 & 0 & \frac{6EI_z}{\ell^2} \\ & & \frac{12EI_y}{\ell^3} & 0 & -\frac{6EI_y}{\ell^2} & 0 & 0 & 0 & -\frac{12EI_y}{\ell^3} & 0 & -\frac{6EI_y}{\ell^2} & 0 \\ & & & \frac{GJ}{\ell} & 0 & 0 & 0 & 0 & 0 & -\frac{GJ}{\ell} & 0 & 0 \\ & & & & \frac{4EI_y}{\ell} & 0 & 0 & 0 & \frac{6EI_y}{\ell^2} & 0 & \frac{2EI_y}{\ell} & 0 \\ & & & & & \frac{4EI_z}{\ell} & 0 & -\frac{6EI_z}{\ell^2} & 0 & 0 & 0 & \frac{2EI_z}{\ell} \\ & & & & & & \frac{EA}{\ell} & 0 & 0 & 0 & 0 & 0 \\ & & & & & & & -\frac{12EI_z}{\ell^3} & 0 & 0 & 0 & -\frac{6EI_z}{\ell^2} \\ & & & & & & & & \frac{12EI_y}{\ell^3} & 0 & \frac{6EI_y}{\ell^2} & 0 \\ & & & & & & & & & \frac{GJ}{\ell} & 0 & 0 \\ & & & & & & & & & & \frac{4EI_y}{\ell} & 0 \\ & & & & & & & & & & & \frac{4EI_z}{\ell} \\ & & & & & & sym. & & & & & \end{bmatrix}$$

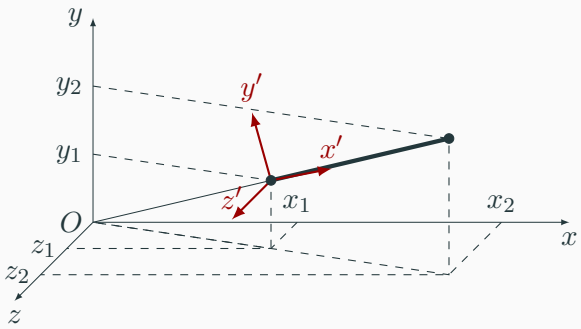
Element consistent mass matrix in local coordinates

The consistent mass matrix for a three-dimensional uniform beam element is written as the superposition of the **axial mass** matrix, the **torsional mass** matrix and the **flexural mass** matrix:

$$\mathbf{M}_{loc} = \frac{\rho A \ell}{420} \begin{bmatrix} 140 & 0 & 0 & 0 & 0 & 0 & 70 & 0 & 0 & 0 & 0 & 0 \\ & 156 & 0 & 0 & 0 & 22\ell & 0 & 54 & 0 & 0 & 0 & -13\ell \\ & & 156 & 0 & -22\ell & 0 & 0 & 0 & 54 & 0 & 13\ell & 0 \\ & & & \frac{140J}{A} & 0 & 0 & 0 & 0 & 0 & \frac{70J}{A} & 0 & 0 \\ & & & & 4\ell^2 & 0 & 0 & 0 & -13\ell & 0 & -3\ell^2 & 0 \\ & & & & & 4\ell^2 & 0 & 13\ell & 0 & 0 & 0 & -3\ell^2 \\ & & & & & & 140 & 0 & 0 & 0 & 0 & 0 \\ & & & & & & & 156 & 0 & 0 & 0 & -22\ell \\ & & & & & & & & 156 & 0 & 22\ell & 0 \\ & & & & & & & & & \frac{140J}{A} & 0 & 0 \\ & & & & & & & & & & 4\ell^2 & 0 \\ sym. & & & & & & & & & & & 4\ell^2 \end{bmatrix}$$

Transformation of coordinates

- The stiffness and mass matrices are defined in the local coordinate system x', y' and z' fixed to the beam segment.
- To assemble the global stiffness and mass matrices, these local matrices must be transformed into the global coordinate system x, y, z.



Direction cosines

- The relationship between local and global coordinates is:

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \underbrace{\begin{bmatrix} \cos(x'x) & \cos(x'y) & \cos(x'z) \\ \cos(y'x) & \cos(y'y) & \cos(y'z) \\ \cos(z'x) & \cos(z'y) & \cos(z'z) \end{bmatrix}}_{\mathbf{T}'} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

- The local x' is given by:

$$Ox' = [l_x, l_y, l_z]$$

where

$$l_x = \cos(x'x) = \frac{x_2 - x_1}{{}^e\ell}, \quad l_y = \cos(x'y) = \frac{y_2 - y_1}{{}^e\ell}, \quad l_z = \cos(x'z) = \frac{z_2 - z_1}{{}^e\ell},$$

$${}^e\ell = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}.$$

Direction cosines

- The local y' axis is chosen so that it lies in the local $x' - y'$ plane and it is perpendicular to x':

$$Oy' = \frac{1}{d}\,[l_y, l_x, 0]$$

thus

$$\cos(y'x) = \frac{l_y}{d}, \quad \cos(y'y) = \frac{l_x}{d}, \quad \cos(y'z) = 0, \quad d = \sqrt{l_x^2 + l_y^2}.$$

- The local z' axis is chosen so that $Oz' = Ox' \times Oy'$:

$$Oz' = \frac{1}{d}\,[-l_x l_z, -l_y l_z, 0]$$

thus

$$\cos(z'x) = -\frac{l_x l_z}{d}, \quad \cos(z'y) = -\frac{l_y l_z}{d}, \quad \cos(z'z) = d.$$

Global displacements for oriented three-dimensional beam

$$\underbrace{\begin{bmatrix} q'_{1x}(t) \\ q'_{1y}(t) \\ q'_{1z}(t) \\ \phi'_{1x}(t) \\ \phi'_{1y}(t) \\ \phi'_{1z}(t) \\ q'_{2x}(t) \\ q'_{2y}(t) \\ q'_{2z}(t) \\ \phi'_{2x}(t) \\ \phi'_{2y}(t) \\ \phi'_{2z}(t) \end{bmatrix}}_{\mathbf{q}_{loc}(t)} = \underbrace{\begin{bmatrix} \mathbf{T}' & & & \\ & \mathbf{T}' & & \\ & & \mathbf{T}' & \\ & & & \mathbf{T}' \end{bmatrix}}_{\mathbf{T}} \underbrace{\begin{bmatrix} q_{1x}(t) \\ q_{1y}(t) \\ q_{1z}(t) \\ \phi_{1x}(t) \\ \phi_{1y}(t) \\ \phi_{1z}(t) \\ q_{2x}(t) \\ q_{2y}(t) \\ q_{2z}(t) \\ \phi_{2x}(t) \\ \phi_{2y}(t) \\ \phi_{2z}(t) \end{bmatrix}}_{\mathbf{q}(t)}$$

